"If I had an hour to solve a problem,
I'd spend 55 minutes thinking about the problem
and 5 minutes thinking about solutions."
— Albert Einstein

Chapter 1

Computational Thinking and Computational Problem Solving

"The quest of THE ALGORITHM"

Summary
In this chapter, we will discover and learn a few of the finest computational thinking techniques in computer science. These elegant techniques are foundational blocks in solving many challenging problems related to computer science. We shall pose a few analytical and computational problems, which will lead the reader towards discovering and appreciating these beautiful techniques. After mastering these techniques students will master how to reason out their computational thinking. We will keep asking ourselves one very fundamental question which is also the central quest of computer scientists i.e. **How to efficiently solve a computational problem?**

Guessing Game (Binary Searching)

Let's play a game. Zuraiz thinks of a number which is in the range from 1 to 1000. Can you guess that number ?

If you ask a kid this question (imagining a number in your mind) - or a class of kids - they will bombard you with so many tries/answers.

Random Tries: Someone may shout, Is it 1? or 21? or 65? or 330? or 500? or may be 999? or 1000? As an adversary, Zuraiz may not even have thought of a number but kept changing whatever number he has in mind, trying to ensure that the guesser will find it extremely hard to win. In the worst case, students may have to take 1000 tries to guess that number.

Now the question is, can we guess this number in fewer tries? What if Zuraiz only allows Yes/No types of questions…?

Primes/Non-Primes: If someone asks, **"Is it a prime number?"** and Zuraiz replies that it is not a prime number. This attempt will reduce the number of possibilities to 832, as the number of primes within the range of 1 to 1000 is 168. See details on this link (https://primes.utm.edu/howmany.html).

Even/Odds: If someone asks **"Is the number even or odd?"** and Zuraiz replies that it is an odd number. This time we have cut down 500 tries (odd numbers) at once but still have 500 even numbers to guess from.

Interval Search: Someone may think of dividing the problem space into intervals of say 100s. For example, a student asks if the number is less than 100. If Zuraiz replies yes then we can guess the number in only a hundred tries (asking about each number from 1-99). If he replies no, then we ask if the number is in the next interval, i.e. less than 200 but greater than 100. If he says yes, then we will be able to guess the number in 101 tries. Ultimately, if the number is between 900 to 1000 then by following this way of searching by range, we will take 109 tries to guess the number which is much better than 500 or 1000 tries.

Claim: Can we do better?
Yes we can. Actually, you have to reduce the steps to solve the problem, What if you first divide the number (in general it should be "range size") by two and then ask if the number is in the first half or in the second half (by asking something like this: "the number is less than or equal to the middle number"?) After that you can again divide it in ranges of 100 numbers at a time and, in the worst case, you can solve it in 105 steps. Now see, you have a systematic way to solve a problem. Divide the number in different parts and then solve it.

Claim: What do you think, can we still do better or not?
Yes we can, we can repeat the step of dividing the range into two parts and select the one for which Zuraiz nods yes. For example: First we take the middle of the range and ask Zuraiz if the number is in the first half or second half. Whatever Zuraiz answers, select that range. Let's suppose that the number is in the second half, now take the middle of the 2nd half and ask Zuraiz again by taking the middle and repeating the same procedure until you find the number or it keeps shrinking until the range becomes 1 number and that is your answer.

Now the question is: with this recursively (repeatedly) halving technique, how many guesses will be needed to find the answer?

DISCUSSION ON LOGARITHM FUNCTION and EXPONENTIAL FUNCTION (D1)

Exponential Growth, and Exponential decay (logarithm) - Arithmetic, Population, Energy

Exponentiation and logarithm are associated with each other just like multiplication and division.

Exponentiation is a <u>mathematical</u> <u>operation</u>, written as b^n, involving two numbers, the <u>base</u> b and the <u>exponent</u> n. Where n is a positive <u>integer</u>, exponentiation corresponds to repeated <u>multiplication</u> of the base b. That is, b^n is the <u>product</u> of multiplying the base b with itself n times.
Similarly if you start from a number N and you keep dividing that number N by b until it becomes 1 that quantity is called $\log_b N$, e.g. $\log_2 1024$ is 10 as $2^{10} = 1024$. Similarly $\log_2 8$ is 3. Similarly $\log_3 27$ is 3, if the number N is not a perfect base of b then the result will be an approximation e.g. $\log_3 28$ is 3.0331....

Both logarithm and exponentiation are related. A beautiful analogy to understand the two quantities is the following. If you start from a quantity 1 and keep multiplying by a constant (let us say 10) then the quantity will increase in the following fashion

$$1, 10, 100, 1000, 10000, 100000, 1000000, \ldots.$$
$$1, 10, 10^2, 10^3, 10^4, 10^5, 10^6, 10^7, \ldots.$$

Just a little fact to remember: the total number of atoms in the known universe is around 10^{82}. Now if you start from a quantity of 1 and and keep multiplying by 10 then in only 82 steps you will reach as big a number as the number of atoms in the universe. To imagine this, let us read a beautiful old riddle related to an ancient King who wanted to reward his minister. The riddle goes something like this,

If a chessboard were to have wheat placed upon each square such that one grain were placed on the first square, two on the second, four on the third, and so on (doubling the number of grains on each subsequent square), how many grains of wheat would there be on the chessboard at the end?

It seems like a very simple problem and one can imagine the number of wheat grains N will look like this: $N = 1 + 2 + 4 + 8 + \ldots + 2^{64}$. Surprisingly this number comes out to be 18,446,744,073,709,551,615.

This riddle is first known to have been recorded in 1256 by Ibn Khallikan.[1] Another version has the inventor of chess (in some tellings Sessa, an ancient Indian Minister) request his ruler to give him wheat according to the wheat and chessboard problem. The ruler laughs it off as a meager prize for a brilliant invention, only to have the court treasurer's report that the unexpectedly huge number of wheat grains would outstrip all the resources of the ruler. Versions differ as to whether the inventor becomes a high-ranking advisor or is executed.

Always remember if there is a quantity which is increasing with a constant rate of e.g. 5% then that quantity can explode very quickly to reach a humongous total. For example, when you see that the population of a country or the energy requirement of a country is increasing with a 5% increase every year then it actually means that within 70/5=14 years the population or energy requirement of the country will become double. Similarly if the rate is 7% per year, that means in just 10 years, the population or energy requirement will become double. For details watch this talk by Al Bartlett Arithmetic, Population and Energy.

Similar to exponential growth, but reversed in direction, is the process of exponential decay. If a quantity is shrinking by some rate then it will take a few steps to be almost finished in a small time. The sequence look like this, if you have some quantity something like N (may be the same wheat source i.e. 18,446,744,073,709,551,615). And some one says that after a while may be a year the quantity is reduced to half then in only $\log_2 N$ (64 steps) the quantity will become just 1. That quantity, the number of steps to shrink to become 1, is called the logarithm of the initial value with base constant (the rate with which the quantity is shrinking).

Now always remember when you read some statements like deforestation is happening with a rate of let us say 5% per year. It means, in just 70/5 = 14 (where does 70 come from ??? Think about that or watch the lecture by Professor Bartlett) years the forests will shrink to half of the forests remains 14 years ago. This is a horrifying statement.

Watch this amazing documentary on two of the most essential and fundamental mathematical functions to understand very important misunderstood challenges the world is facing related to population, crisis of energy and resources equilibria.

Lecture by Professor Bartlett: http://topdocumentaryfilms.com/arithmetic-population-and-energy-lecture/
http://www.albartlett.org/presentations/arithmetic_population_energy_transcript_english.html

Now if the number is like K (between the range 1 to N) and we have to find this number, this halving technique will take $log_2 N$ steps on the worst case until it finds that number (i.e the interval length reduces to 1 only).

EXAMPLE:

Let's consider that zuraiz thinks of a number 571. Now first we divide 1000 by 2 which is 500 and ask if this number is greater than 500 or not or is it in the first half or second half?

As 571 is in the second half so we will find the middle of the second half (by adding the two extremes and divide by 2 i.e. $\frac{(501+1000)}{2} = 750$) and ask whether it is greater than 750 or not?. As it is not so we will divide this half again by 2 again ask if this number is greater than 625 or not And so on… Here is the table of the questions and the halving range with respect to that the guessed range is shrinking slowly.

STEP #	RANGE	GUESSED NUMBER	Answer	Shrinked Range
1	1-1000	<501	No	501-1000
2	501-1000	<751	Yes	501-750
3	501-750	<626	Yes	501-625
4	501-625	<562	No	562-625
5	562-625	<594	Yes	562-593
6	562-593	<578	Yes	562-577
7	562-577	<569	No	569-577
8	569-577	>573	No	569-573
9	569-573	>570	Yes	571-572
10	571-572	>571	No	**571 (is the answer)**

So the Number is 571.

EXERCISE

Q1. Can you tell that in how many steps we can find the same number if our range is 4000?

Q2. If the range is of N size from 1-N it will take approximately $log_2 N$steps. Prove it? Read Section D1 for directions.

Finding The Heaviest Ball

Let's suppose that we have a balance and 8 metal balls which are equal in weights except for one which is heavier than all. Now we have to tell which one is the heaviest ball? The only way you can compare the weight of the two balls is by placing them on the balance and checking which one bends down.

[Image from] http://mkcohen.com/wp-content/uploads/2011/03/scale.jpeg

- ➔ We can find the heaviest ball by placing one ball on the left side and putting all balls one by one on the other side of the balance (the side which goes down has the heaviest ball). This method will take at most 7 tries to get the heaviest ball.
- ➔ If we have N balls then it will be very difficult for us to find the heaviest ball (N-1 Tries at max).

Can we make it in less tries?

We can find the heaviest ball by distributing balls into two parts i.e., putting half balls on one side of balance and half on the other side and check which side of balance is heavy; we throw all the balls on the lighter side (as they can't have that heavier ball) and select the one which is the heavier side and then apply same halving technique (as of Ref: Guessing Game) to the heavy set of balls i.e., dividing them into half and put each half onto each side of balance and check which side is heavy and continue with the heavier one till we get the heaviest ball.

This technique will take $log_2 N$ solution (**log** is described above DISCUSSION ON LOGARITHM FUNCTION (D1) so you can consult with it).

Is this the smartest way? Hint: Can we solve the 3 balls problem in One try?

Yes, put one ball on one side of balance and another ball on the other side of the balance and the third ball outside of the balance, Now we can determine the heaviest ball among three balls in one step, if the heaviest ball is placed on the balance then that side will bow down and if balls on balance are equal weighted then we can say that third ball placed outside the balance is the heaviest one (given that "*one of the ball was heavier*").

Can we now solve the 8 balls problem in 2 tries?

Yes, we can find the heaviest ball in 2 tries by placing 3 balls on the left side, 3 balls on the right side and 2 balls outside the balance. Now if the left side bow down it means the the ball is in left side (we reject the other

3 and the 2 placed outside). Now we can solve this by the method described in the above paragraph, the 3 balls problem in one try.

Similarly, Let's suppose that we have 9 balls then we can apply the same scenario by putting 3 balls on the left side, 3 on the right side and three are outside of the balance. If it is balanced then we will put one ball on the left side (The balls that were outside of the balance) and one on the right side and one is outside of the balance (except same scenario as above). This method will again take 2 steps to find the heaviest ball.

EXERCISE

Ex 1.3.1 Solve the same above problem for 15, 25 and 27 balls in just 3 tries.

Ex 1.3.2 Try Solving 28 with 3 tries only, You won't be able to solve it?

Ex 1.3.3. Solve the 64, 81 balls case in 4 tries.

Ex 1.3.4. In general if we have N balls then we will do the same procedure (by putting $\frac{N}{3}$ balls On the left side, $\frac{N}{3}$ balls on the right side and $\frac{N}{3}$ outside the balance). Now the question is how many tries will it take to find heaviest ball from N balls We can calculate this by using the formula $log_3 N$. Hint: Read D1.

Ex 1.3.5. What will happen in case N is not completely divisible by 3?

The Proposal Story (Sarim-Katrina ring exchange)

Once there were two love birds Sarim and Katrina. Sarim lives near the river and the girl Katrina lived across the river. They send messages through the river by placing the letter inside the box and sending it through the river (which you can safely assume takes the box to both ends). They both love each other. Sarim loves Katrina and wants to propose to Katrina by sending the ring to her through the same communication channel, river. Sarim is facing a huge problem, as the river is being attacked by the pirates and they check every box (if it is unlocked) and see if it has a precious thing they steal. If it doesn't then they let it go (also if the box is locked they let it go too).

Sarim and Katrina have boxes and locks.
Now you have to think of a solution by which Sarim sends the ring to Katrina easily.
Note: Sarim and Katrina have many boxes and locks along with their keys them. As shown in the figure.

What do you think?

If Sarim places the ring in the box, locks the box and sends it to Katrina. He has successfully saved the ring from the pirates and Katrina collects the box of the ring successfully but wait do you think that the problem is solved here and *Katrina has got the ring?*
No. Actually, Katrina got the locked box of rings, not the ring. Hence our problem is still not solved.
What if She again locks the same box (double lock) and sends the box toward him and he unlock his own lock and again sends the box toward her, Now the box has only one lock which is of Katrina's lock and she can easily remove the lock from the box because lock on the box was her own and obviously she has the key of that lock. She unlocked the box and then took out the ring from the box. Viola, the problem is solved, Proposal accepted

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Cryptography
Modern social and web-based online world works under the the principle of cryptography which is derived under the very fundamental requirements i.e. all the information (called messages or information data packets) shared/send over the internet are encrypted in some form which if anyone tries to hack then one shouldn't be able to understand what the message means. Only the receiver of the message should be able to understand what the message actually means. The data passes through the internet network (is like a river in the above example) and reaches to the destination servers (just like the two ends of the rivers), pirates are the access points where anyone connect (the internet providers with privileges of course) and can hook the network packets passing through the internet connections. All the encryption versions of messages (i.e. ring inside the locked box of the above example) passed over the internet works in the similar fashion as stated on the above example, actually in a bit simpler fashion.

For example a very powerful encryption method called RSA Encryption Protocol is used by most of the giants of social world enterprise businesses like Facebook, Google, and so on. RSA Encryption Protocol works in a following fashion. If an end user U wants to send a message M to server S. What our social server S does is that it publishes on its server its lock L (which is like a padlock in the above example and that lock is send across all the corners of the river hence everyone has that lock of S) and it is available to everyone in the world including the end user U. The key K of the lock L is only there with the server S which is hidden. What U (in the above case Sarim) does is using L (padlock of Katrina) he encrypts M (Sarim's ring) and make M' (ring in the box locked with the lock of Katrina) by taking some mathematical operations (like some multiplications, divisions like arithmetic operations) which is quite easy to compute, if you know M and L, and send it over the internet. When the encrypted message M' reaches the server S (i.e. when locked Box containing the ring reaches to Katrina), to recompute M from M' there is only one practical way to figure out what is M i.e. by using L (just like in above example the padlock was only possible to open through the keys of those who is the owner of the lock). Without using L it is possible to find M from M' but the only possible way is trying every possibility which will be extremely impractical i.e. like the solution will have exponentially many possibilities which even a supercomputer may take millions of years to crack.

If you are interested See details from here.
https://www.quora.com/What-is-public-key-cryptography-in-laymans-terms-How-is-it-related-to-prime-numbers
https://www.quora.com/What-is-a-general-overview-of-the-RSA-cryptography-in-laymans-terms
http://sergematovic.tripod.com/rsa1.html
RSA link: BOOK's link

Counting the population of the country (Exponential growth and Parallelism)

Problem: How would you count the number of students present in the class?
One Naive idea could be that we give a task to one person to see through all the people step by step and keep counting every person. This procedure will take N steps where N is the number of students in the class. Let us call this idea IDEA 1.

Suppose we are in a Hall now and it is full of an audience and we want to calculate the number of people N (for example close to 1000) in the stadium. How will you count them?

Now as we know that the audience is well seated in a uniform arrangement of seats, i.e. each row has an equal number of people in it. We can count in a much efficient way as compared to the IDEA 1 (which will take 1000 steps or in general N steps). For that, we will count the number of people in one row, let us say there are C-people, and then count number of rows in the hall, let us say that we have R many rows then total number of people will be rows count multiplied by number of people in one row i.e. C x R. The important question is that "Is this idea better than IDEA 1?". In this idea, the steps counts (counting and additions and multiplications) will be C+R+1 (C is for counting a row size, R is counting for number of Rows and 1 is finding the total number of people by multiplying R and C. Let us call this idea IDEA 2.

Let us visualize this IDEA 2 with a proper example.

- Now let us say we have counted the number of rows and they appeared to be 10 rows and we have also counted the number of people in one row and they appeared to be 100 people. Now the number of people sitting in the hall will be equal to 10x100 = 1000 people. Now we want to count that it took us how many steps to calculate the number of people in the hall. For that, as we have derived a formula which is C+R+1,(In this particular case our R is 10-rows then C will be 100 (number of people in one row) and the total step count will be 10+100+1= 111).

- Similarly, If I have to make the solution of the same scenario as above but with 20 rows and we have also counted the number of people in one row and they appeared to be 50. Now the number of people sitting in the hall will be equal to 20x50 = 1000 people. Now we want to count that it took us how many steps to calculate the number of people in the hall. For that, as we have derived a formula which is

C+R+1,(In this particular case our R is 20-rows and C is 50 number of people in one row) and the total step count will be 20+50+1= 71).

Exercise

Ex 1.5.1: What will you do if the first row of the hall has 40 people and there are 25 rows. What will be the number of people in the hall? It will take how many steps to find the number of people in the hall?
Ex 1.5.2: What will you do if the first row of the hall has 50 people and there are 20 rows. What will be the number of people in the hall? It will take how many steps to find the number of people in the hall?
Ex 1.5.3: What will you do if the first row of the hall has 100 people and there are 10 rows. What will be the number of people in the hall? It will take how many steps to find the number of people in the hall?
Ex 1.5.4: On what values of R and C, Can you suggest a way by which we can calculate 'N' number of people in the minimum number of steps?
Hint: Look at the pattern in the above three exercises (it first decreases as the first row count is increasing and number of rows decreases and then it starts increasing again)?

Parallelism

Counting the crowd size in a circular stadium

Let us suppose that now we have a crowd sitting in a stadium and we want to count the number of people in the stadium. It Is obvious that the smart IDEA 2 will fail here, because the circumference of the inner circle is much smaller than the outer circle and the capacity increases as we move from inner circle to outwards. Similarly IDEA 1 will be extremely time taking. So we have to come up with a smarter new idea.
The new idea is that we will count each row simultaneously at a time. As maximum people are in the outermost circle so all the people in other circles will be counted when counting of this circle will complete. So the total cost for counting all the people in the stadium will take counting the outermost circle and the total circles in the stadium.

Counting the population of the class and generalizing it to the country's population?

For that, to count **N** number of people in the class we will first ask everybody to take out the paper and then at the very upper left corner we will write our turn number (how many times you got the paper) and then at the lower right of the page we will just write the total number of students. At the very first instance you will see that you haven't got any paper, so you will just write zero on the upper left and 1 on the lower right because you are single. Now we will ask the whole class to stand-up at once and make a pair of two, at that moment each group has 2 pages with them(each person has one page).Now we will make one representative from each pair and then ask the other person to sit down and give his paper to his group partner, and then ask the representer to write the 2nd turn on upper side and in the middle write the the summation of the two numbers of both the papers that will be two for each representative. At that instance you will see that the representer has two pages (his own and his group partners page), you will write **1** on upper left because for the first time you got the paper and **2** on the lower right because you are in a pair with a person. Now for the second time

we will again make the pair but at that time we will make the pair of just representatives (people who are still standing), and ask one of them to be again a representative and the other to sit down and again as earlier representatives will write the numbers on the left upper and lower right. We will repeat this step until there is only 1 man standing. Now at last you will see that at upper left you have a total number of steps to count the people and at lower right you will see the total number of people which is **N**.

If you look at this idea carefully this beautiful solution is extremely powerful and generates Parallel counting the crowd size in a stadium. Here instead of 1 person for each row everybody contributed in counting the size of the population. The people's standing will be the active representatives and holding the summation of the population count to whom they represent. The count will look like in the following fashion.

At the very beginning N people will be standing having a page on which there is written ${}^{0}P_{1}$. After the first step N/2 people will sit down and N/2 will become the representatives having their page having the number ${}^{1}P_{2}$. After the 2nd step every N/4 person will be standing having written pages ${}^{2}P_{4}$. After the 3rd step every N/8 people standing will have ${}^{3}P_{8}$ written. Now the question it will take how many steps to find the exact count of the population. If you look at the shrinking sequence of representatives it will look like this,

$$N, \frac{N}{2}, \frac{N}{4},, \frac{N}{8},, \frac{N}{16}, \frac{N}{32}, \ldots\ldots, 1$$

It will take $log_2 N$steps to count exactly what is the population of the country. If you look at the population right now i.e. 7.5 billion on 11th Aug 2017 to be approximately 7,524,594,281 for this much value of N the step counts will just be approximately 33 steps ONLY. That is the power of parallelism.

REF: http://blogs.dunyanews.tv/wp-content/uploads/2017/06/Pakistan-India-final.jpg

Thus by this way (DISCUSSION ON LOGARITHM FUNCTION (D1)) we can count elements in universe only in 150 steps, thus this is the power of parallelism.

Chapter 1: Challenges

(Assignment # 1)

Challenge 1.1

1. You have 2 drums of capacity
   - ➔ 8 litres
   - ➔ 5 litres

And a resource of Oil, Can you use these two drums to give 1litre, 2-litre, 3 liter, 4 liter, and 6 liters and 7 liters to any customer. Note that there is no mark of liters on the drums.

2. Now you have to measure again the same question: But you have a 9-litres and 6 liters drum. Can you solve that same problem, if yes how? If no why?

Directions to look at of Problem 1.1

For example you can easily measure 3 liters in the following way; take 8 liters drum, let us call it A and pour it into 5 liters drum, let us call it B, until B gets filled. Now the remaining liters in A will be 3. Similarly, repeat the same procedure (you can throw the filled drum again into the resource of Oil).

Challenge 1.2

On your first day at university, the teacher suggested that it would be a good idea for each student to meet every other student in the class. The teacher said, "When you meet, please shake hands and introduce yourself by name." If there were 20 students in the class, how many total handshakes happened? What about if there are N students can you tell in terms of N what is the total count of shake-hands.

Directions to look at of Problem 1.2

One student will hand-shake with other student only once. Let say if we have 3 students A, B ,C then A student will hand-shake with B and C and B will hand-shake with C only because B has already have a hand-shake with A. Now C will not hand -shake with A and B because C has already have a hand-shake with A and B.

Challenge 1.3

If you have an 8 x 8 chessboard, how many squares (within that chessboard) would there be in total? (64 is not the answer).

Directions to look at of Problem 1.3

You must calculate each possible combination of squares on board i.e., 1x1, 2x2,...,8x8 have how many square boxes and then sum up these # of boxes.

Challenge 1.4

You are given 2 balls made of clay. You have access to a 100-story building. Balls can be very hard or very fragile means it may break if dropped from the first floor or may not even break if dropped from the 100th floor. Both balls are identical. You need to figure out the highest floor of a 100-story building a ball can be dropped without breaking. The question is how many minimum drops do you need to make to find that story. You are allowed to break 2 balls in the process.

Directions to look at Problem 1.4

First, try to solve the problem with only one ball. And to solve with one ball you can't solve that with binary solutions like if you drop that from the 50th floor and it breaks then you are bound. So, for one ball you can only solve that by dropping the ball from the 1st floor than from the 2nd and so on. But in that case, we have a lot of steps to solve. Like if You have an N floor then you have N-1 tries.

Now as you have two balls to solve that problem, you can first drop that from the 50th floor and if it breaks then you can use the same solution which we used in case of one ball from the 1-49th floor and if it doesn't break then you drop the second ball from 51 to 100th floor. Now see that, to figure out the solution with one ball we have done

99 tries (worst case). But in that case with two balls we will just try 49 times to solve the solution in worst case. **Now the question is that can you do better than that?**

Challenge 1.5

Divide RS: 277 (in whole $ increments) into a number of bags so that I can ask for any amount between RS: 1 and RS: 277, and you can give me the proper amount by giving me a certain number of these bags without opening them. What is the minimum number of bags you will require?

Directions to look at of Problem 1.5

You must derive a sequence like if you make a bag having RS: 1 and RS: 3 then there is no need to make a bag of RS: 4 again because 4 can be paid using RS: 1 and RS: 3 bag combined. Similarly try different possibilities to figure out what should be the minimum sequence to have so that every amount from 1 to 277 can be paid with the subset of the set of bags you will make.

Challenge 1.6

A corporate businessman has two cubes on his office desk. Every day he arranges both cubes so that the front faces show the current day of the month. What numbers are on the faces of the cubes to allow this?

Note: You can't represent the day "7" with a single cube with a side that says 7 on it and the other cube hidden. You have to use both cubes all the time. So the 7th day should be represented like "07".

Directions to look at Problem 1.6

You must detect that numbers which can represent every day of the month and a number on one cube can be on another cube but repetition of numbers on the same cube is not allowed. One cube can have a series like 0,1,2,3,4,5. Now you have to detect the numbers on second cube.

Challenge 1.7

A pot contains 75 white beans and 150 black ones. Next to the pot is a large pile of black beans.

A somewhat crazy cook removes the beans from the pot, one at a time, according to the following strange rule: He removes two beans from the pot at random.

- If at least one of the beans is black, he places it on the bean-pile and drops the other bean, no matter what color, back in the pot.
- If both beans are white, on the other hand, he discards both of them and removes one black bean from the pile and drops it in the pot.

Notice at each turn of this procedure, the pot has one less bean in it. Eventually, just one bean is left in the pot. What color is it?

Directions to look at of Problem 1.7

1. black black beans: (-1) black bean
2. Black white beans: (-1) black bean
3. White white beans: (-2) white (+)1 black

This is the pattern of beans are added back to the pot or removed from it. Now determine at end which bean will be in the pot according to these rules. *Try solving with a small example with 5 white beans and 10 black beans with random choices of pairs*.

Challenge 1.9

You have an infinite plane and lines, lines can intersect each other to make regions, e.g. 1 line will produce 2 regions, 2 lines 4. Can you tell with argument at maximum how many regions will be there if you have n lines? And why? (Any answer without reason will be unacceptable).

Directions to look at of Problem 1.9

From above we can determine that 1 line produces 2 (i.e., 1x2) and 2 lines produce 4 (i.e., 2x2), with three lines you can have at max 7 regions. Now determine a generalization for N lines what my number of regions will be. Look very carefully at what happens with the introduction of a new line and how it will make the maximum number of regions? Use the help of drawing.

Challenge 1.8

You wake up after your Titanic sank, and you find yourself on the shore of the sea, an island. Besides everything, that an island has (its natural beauty), there are tribesmen there too. You come to know that every tribesman of this island is either faker(he may or may not lie) or trustworthy(always say the truth). You can ask them questions like who is a faker. And who is trustworthy? There are a total of 100 people on this island. And you can ask as many questions as you want. The only information known is that the number of Trustworthiness is greater than fakers. Any answer having greater than 500 questions is not worthy enough to be considered. Figure out who is a faker and who is a trustworthy person?

The Geometric Series

Geometric Series is one of the most essential and exists in abundance in nature and also in computational related problems in CS. It looks like the following:

$1 + b + b^2 + b^3 + b^4 + \ldots + N/b^2 + N/b^1 + N$ or also written as $N + N/b + N/b^2 + \ldots + b^4 + b^3 + b^2 + b + 1$

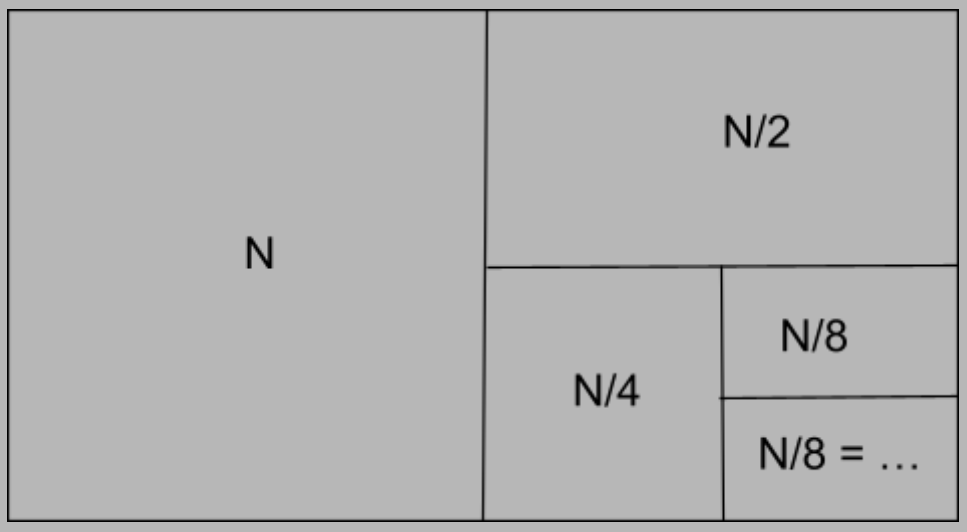
The left side is known as increasing geometric series and the right side is decreasing geometric series both ending and starting with N.

It is not hard to prove that that the summation of the above series is bounded by a constant time the largest term i.e. some C x N. See Wikipedia for the proof **https://en.wikipedia.org/wiki/Geometric_series**

Example of the above series could be the following series with b = 2.

$1+2+4+8+16+\ldots+ N/4 + N/2 + N$ also written as $N + N/2 + N/4 + N/8 + \ldots + 16 + 8 + 4 + 2 + 1$

It is not hard to prove that the summation of the above series (2 versions increasing as well as the decreasing geometric series is always less than 2N. The proof is quite simple and very simple to figure out what it means. Imagine that the largest term N is actually one big Square S take two big squares now the claim is the rest of the summation can never together sum up and fill in the 2nd square. Very simply start shading the summed up region. If N is a complete square S then N/2 will be half of the square. Similarly N/4 will be a quarter of the square S and so on. Hence you will never be able to fill in the 2nd square S. That completes the proof that the summation of the geometric series with increasing or decreasing factor b=2 is always bounded by <2N.



Directions to look at of Problem 1.8

You have to find at least one trustworthy person by whom you can ask who is a trustworthy or liar. The strategy is inspired by the beautiful Geometric series. One idea could have been to take one person out and ask from everyone "Is he the trustworthy one?" if the majority say yes then we have found the one trustworthy and we may ask him about everyone. But the issue is you could be unlucky in that case you have to choose the 2nd person and repeat that would be a very bad solution. It will work but in that case you may have to almost select 49 bad choices of fakers.

Now the idea is make pair of 2 each (resulting 50 pairs i.e. N/2 pairs) and just ask those pairs about each other Here could be the possible answers (F for faker, T for Trustworthy).

Case 1	F	F
Case 2	T	F
Case 3	T	T

In all those cases where cases 1 and 2 arise we reject the pairs. In case 3 we choose one representative and ask the other person in the pair to sit down. With this one process greater than half of the people have been pruned and you should prove that "the Trustworthy-men in majority" condition will still be valid (you should prove that?). With this one step we have asked 100 questions but the searching space have reduced to half, now we repeat the same process of shrinking and on the worst case the number of questions will look like the following series 100 + 50 + 25 + … + 1 that will always be less than 200 questions. Hence we have found one Trustworthy in around 200 questions.

Be careful with this procedure. It may be possible that there is an odd number of people remaining on a certain step. What should we do in that case? We have to think about that. The hint is in incorporating that issue we might have to do some extra steps which might take the number of questions close to 300.

Challenge 1.10

- A duck is pursued by a fox, escapes to the center of a perfectly circular pond. The fox cannot swim and the duck cannot take flight from the water. The fox is 4 times faster than the duck. Assuming the fox and duck pursue optimal strategy, is it possible for the duck to reach the edge of the pond and fly away without being eaten? If so how? (You can find its answer on the Internet very easily, you can use the Internet but I would say first try for it and then go on finding it) [2]
- On what parameters (relation between their speeds) fox will always be able to eat it, and on what factors duck will always escape.

Directions to look at of Problem 1.10

Use the freedom of the movement of the duck inside the pond such that the duck can have the prior distance covered already so that when the duck decides to move in a certain direction of reaching to the corner such that when it reaches and puts a foot on the ground fox is still lagging in the chase.

Challenge 1.11

You have two strings whose only known property is that when you light one either end of either string it takes exactly one hour to burn. The rate at which the strings will burn is completely random and each string is different. How do you measure 1.5 hours and 45 minutes?

Directions to look at of Problem 1.11

In order to measure 1.5 hours you firstly burn a string from one side and when it burns completely 1 hour is passed then simultaneously burn second string from both sides and when it is burned completely 1.5 hours are gone. Now use a similar strategy to calculate 45 minutes.

Challenge 1.12

There are four numbers: 1, 2, 5, 10 that you have to carry in a boat across a river. Every time you may not carry more than two numbers , and the Cost of traveling is associated with the largest of two numbers. One number has to be in the boat

during the return journey, which also defines the Cost for this trip. What is the minimum total Cost to carry all numbers from one bank to another?

Directions to look at Problem 1.12

If we take the number 1 and 2 to cross the river or take back the number 1, then it will cost us 2+1 = 3.

Then we take number 5 and 10 and return back with 5 (10+5 = 15). And at last we take 5 and 1 across the river that makes it total cost of 23. **Can we do it in a better way?**

Challenge 1.13 (Exponential Search)

Let us play a game. This is a One-Dimensional game in which we have many people/lives(**N** people) in a single row. Accidently these people/lives are affected by a deadly virus. This virus spreads to people one by one in a specific order and maybe when we start the game then there are already many people/life affected by the virus but fortunately, Dr. Sarim made the antidote for that virus. As we know, viruses affected people continuously. So, when we check any life for a virus that affects that life or not, then it will take a little bit of time but during that time the virus will affect another life who will be next to the previous affected life. We can check and give an antidote to people/life one by one, meaning only one life will be checked and get antidote at a time. So, we need to save the maximum number of people from the virus. How many minimum people will be affected by the virus during finding last affected by the virus?

Direction to look at Problem 1.13

Suppose there are 100 people/life in this game and we need to save maximum people from the deadly virus. First, we will check mid-life which is 50th life, so we will check whether 50th life is affected or not. If affected then we will check for mid of 50 & 100 which is 75th life. But if 50th life is not affected then we give him an antidote and check for mid of 0 & 50 which is 25. So, by that process, we can save 94 people/life out of 100 people/life. But what about we have **N** people/life. You have to save as many people/life as possible from the deadly virus.

References

[1] https://www.clsbe.lisboa.ucp.pt/sites/default/files/styles/landscape_16_9/public/assets/images/-l6b5145.jpg?itok=ZtRxuMZT

[2] Cracking the Code Interviews Book